

# A Mostly Harmless Introduction to Claude Code, Codex and Friends For Empirical Researchers

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# About me

## Daniel Sánchez, MA

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## Working with agentic AI

- Design GenAI workflows (Claude, ChatGPT, Gemini, Copilot) for economic research/analysis
- Developing Model Context Protocol servers for data access with agentic AI assistants
- Training researchers and analysts on agentic AI for different purposes

# Agenda

- 1 Introduction: what AI is, and why agentic AI
- 2 What's out there: tools and models
- 3 Before we get started: cybersecurity basics
- 4 Usage
- 5 Cybersecurity: local models

# AI, anyone?

- What people call “AI” today mostly means one thing: large language models (LLMs), not the general concept of “AI”
  - technically, we’ve had AI for a while (ML, even regression)
  - but the current wave of AI is powered by LLMs
- An LLM is trained to predict the next word from huge amounts of text and code
  - hence the “token” talk: tokens are groups of words
- Three things had to come together for that to work:
  - Data: decades of text and code, digitized and sitting on the internet
  - Compute: GPUs cheap and parallel enough to train on data at that scale
  - Architecture: the Transformer (2017), which made training at that scale practical

# A word on tokens

- Usage is metered in *tokens*: chunks of the text a model reads and writes
- Every prompt and every response consumes tokens
  - Big tasks (long documents, large codebases) consume more
- Subscriptions (e.g. Claude Pro/Max) come with usage caps; the API instead charges per token, pay only for what you use

# Usage limits: can you have your cake and eat it too?



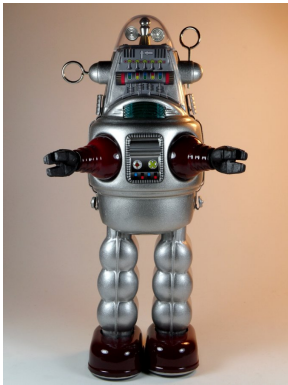
- The AI company gives you one cake for the week
  - Bigger subscriptions get bigger cakes: typical tiers run about \$20, \$100, or \$200 a month
  - Every prompt costs tokens: that's you taking a spoon and getting bites of the cake

# Usage limits (cont.)

- Most companies also cap you at 5-hour windows
  - Instead of letting you eat straight off the cake, they cut it into slices, one slice every 5 hours
  - Genuinely, the most annoying limit
- Companies differ in how permissive they are:
  - Anthropic: not permissive at all; they don't own their own data centers, so it's a markedly more expensive subscription
  - OpenAI: lets you eat straight off the cake, no 5-hour slicing, but only on the most expensive subscriptions
  - Cursor: no slicing at all, you can eat the whole cake in one day if you want
- You can top up with extra usage, but it costs more per bite; past a point it's worth moving to a bigger subscription instead

# Vanilla AI vs. agentic AI

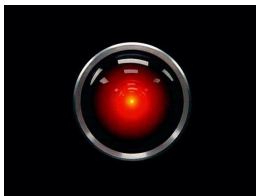
## Vanilla AI (ChatGPT, Claude.ai, a chat window)



- You paste in code or data, it replies with text
- You copy the suggestion back out, apply it yourself, rerun it yourself
- It never sees whether its own suggestion actually worked
- We've had it for a while, but at a certain point, it stops being *that* useful

# Vanilla AI vs. agentic AI (cont.)

## Agentic AI (Claude Code, Codex, and friends)



- It gained *agency*: acts on its own, not just replying to you
- It opens your files, makes the edit itself, runs the script itself
- It reads the real error or output and tries again, without you relaying it
- The unit of delegation moves from *a question* to *a task*
- What made this possible: larger context windows and reliable tool use, not simply “a smarter model”

# Why even care?

- Point the AI to your paper folder: the sky's the limit
- Read your raw data, write and run the cleaning script, and notice when a value label silently changed across survey waves
- It can run your regression, read the actual error message, and fix the bug, instead of you pasting the traceback back and forth

# What it does **not** do

- It does **not** replace your judgment on identification, specification, or interpretation: you still decide what's correct, it does the retyping and rerunning
  - Many times, they hallucinate a solution that looks plausible but is actually wrong
  - No context or economic intuition: it's a *stochastic parrot*, it cannot think
- Net effect: fewer hours lost to mechanical iteration [writing every code line by line], more time on the parts of the research that actually require you
- It doesn't know what cybersecurity or data privacy rules apply to your data, so you still have to be careful

# What's out there

- Terminal / CLI agents: Claude Code (Anthropic), Codex (Open AI), Cursor (now SpaceX), GitHub Copilot (Microsoft), etc.
  - Each of these has its own desktop program version and/or IDE plugins
- Non-technical agentic software:
  - ChatGPT Work
  - Claude Cowork
  - Copilot Cowork

# And what about the “models”?

- AI bros increasingly talk about “models”
- In practice, the model is the “engine” that powers the agent
- Anthropic: Haiku (fast), Sonnet (default), Opus (powerful)
- OpenAI: Luna (fast), Terra (default), Sol (powerful)
- Gemini: Gemini fast, Gemini pro
- Grok, Kimi, Deepseek, so many more...

# Before we get started

- Cybersecurity is a real concern:
  - Secret/private data with sensitive info (e.g. social security numbers) shouldn't be used into an agentic AI
  - For this, there's *local* models, which we may discuss later,
- Don't paste stuff from webpages you don't trust into the agent, it may be malicious
  - Prompt injection: trick your AI into doing something you didn't intend
- An agent can delete your whole project folder if you tell it to
  - Generally they have a "classifier" which tells them whether a command is safe or not, but it can be tricked
  - Use Git!



# Usages: vibe coding a data cleaning script

- Vibe coding: coined in 2025, means prompting the agent to write code while you watch and comment
  - The agent writes the code
  - You provide the vibes
- Case study: 12 files of employment survey data from Ecuador
  - We need to clean, select the right variables, and reshape it into a province-month panel

# Usages: running some regressions

- The agent can run the regression for you, and help you interpret the output
- You can even use it for multiple specifications
  - Maybe you didn't have time before to code so many of them, now you do it in less than 10 mins

# Usages: code review and file organization

- One of the most painful things ever is reviewing a coauthor's code or some old code you wrote a year ago
  - The folders are a mess
  - The comments are terrible
  - You never wrote documentation, lol
- An agent can help you out by
  - Reviewing the code and suggesting improvements
  - Suggesting a better folder structure (and implementing it)
  - Writing documentation for your code for you (or for another agent)
- Case study: my own replication package for Holcombe & Boudreaux (2015), *Regulation and Corruption*, and analyze

Holcombe, R. G., & Boudreaux, C. J. (2015). Regulation and corruption. *Public Choice*, 164, 75-85.

# Usages: code review and file organization (cont)

- Example prompt to ask an agent to check a paper's methods and code:

*"Evaluate the main results in the paper. For each result, ask: Is the method conceptually sound? Does the code correctly implement the method? Make sure to point out anything that could be a fatal flaw; e.g. if the code works assuming the panel is balanced, but fails when the panel is unbalanced, then flag this as a serious problem, because panel data is often unbalanced."*

- Source: Michael Wiebe, *Can AI do replications? GPT5.2 vs GPT5.4 vs refine.ink.*  
[blog.michaelwiebe.com/p/can-ai-do-replications-gpt52-vs-gpt54](https://blog.michaelwiebe.com/p/can-ai-do-replications-gpt52-vs-gpt54)

# Usages: help your agent help you

- Agents love markdown format instructions
  - Markdown:
    - Use headings to separate tasks
    - Use bullet points to list instructions
    - Use code blocks for code snippets
- You can ask your own AI to see your messy code/project, and write an AGENTS.md file
  - Tells what the agent should do, and how to do it
  - With Claude Code, AGENTS.md needs to be called CLAUDE.md
- Write READMEs, replication manuals, etc. etc. etc.

# Usages: literature review

- Review an existing literature review
- Look for similar papers, and summarize them
  - best done using the “deep research” capability of agents
- Use connectors/MCPs to investigate specific literature databases
  - paper-search-mcp: MCP, CLI, and Skills in one; searches and downloads from arXiv, PubMed, Google Scholar, Semantic Scholar, SSRN, and more
  - scholar-mcp: queries Google Scholar directly, pulls metadata, citation, and author info
  - paperclip: full-text retrieval from arXiv, OSF, and OpenAlex, converted to clean markdown for analysis

# Usages: writing review and SKILL.md

- Arguable the worst ability of LLMs is writing a good paper
- However, one can create a reproducible skill to teach the LLM to write like yourself
- This is a skill:
  - A reusable prompt that can be used to write a paper in your style
  - Can be shared with other researchers, and they can adapt it to their own style
- Ask the agent to look at your old papers, draft a SKILL, and learn it
- My own example: `daniel-voice`, built from three of my own papers
  - Two registers, academic prose vs. everyday correspondence, picked automatically from the type of output requested
  - Encodes concrete tics: no em dashes, hedge interpretation but never methodology, always name a limitation near the claim it limits
  - [github.com/dsanchezp18/ai-configs-daniel](https://github.com/dsanchezp18/ai-configs-daniel)

# Usages: skills and agents from modern researchers

- A few economists have already started to create skills and agents for their own research
- Paul Goldsmith-Pinkham (Yale SOM): writes and teaches on this directly
  - Runs “Using AI in Research and Teaching,” a course at Yale SOM
  - Substack series on building and using SKILL.md skills for research
- Scott Cunningham (Baylor, *Causal Inference: The Mixtape*): publishes actual skills, not just posts about them
  - MixtapeTools: a public repo of skills (coding audits, deck-building, bib-checking, and more)

# Usages: skills and agents from modern researchers (cont)

- Pedro H.C. Sant'Anna (Emory, of Callaway-Sant'Anna DiD fame): a ready-to-fork Claude Code template for academic LaTeX/Beamer + R projects
  - Multi-agent review, quality gates, replication protocols built in
- Aniket Panjwani (*The AI Economist* newsletter): good intro videos on YouTube, interesting/good tweets on AI workflows, etc.
  - aieconomist.io
  - Emergent Ventures grant (2025) for EconNow, an AI data platform for economists

# Usages: what do Sant'Anna's example skills do?

- Seven of his skills, copied into this course's repo: `examples/santanna-skills/`
- `replication-package`: assembles a submission-ready deposit (AEA DCAS / openCPSR standard): README, dataset manifest, table/figure to script:line map
- `audit-reproducibility`: numerically checks a paper's reported numbers against what the code actually produces, PASS/FAIL per claim
- `respond-to-referees`: drafts a response letter, matching each referee comment to the specific revision that addresses it
- `review-paper`: a full manuscript review, with an optional simulated peer-review mode (editor plus two referees plus a decision)
- `lit-review`: structured literature search and synthesis, with citation extraction and gap identification
- `stata-replication`: scaffolds and runs a full Stata `.do` pipeline via `stata-mcp`, producing publication-ready tables and figures
- `data-management-plan`: drafts a funder-compliant Data Management Plan (NSF, NIH, ERC, Horizon Europe) for a grant proposal

# Usages: MCP for data access

- Ever had ChatGPT complain because it can't get you a specific datapoint?
  - Either it's private, or it's not easily reachable by an agent
- MCPs are add-ons that allow agents to access specific data sources, like APIs or databases
  - E.g. MCPs to access government data, or to access specific academic databases (e.g. JSTOR, Scopus, etc.)
  - Or MCPs to allow the agent to work more easily with other agents, (stata-mcp)
- EcuDataMCP: allows to easily consult the Ecuadorian government database
- Similar MCPs exist for other countries/regions

# Usages: MCP for data access (cont)

- Canada: mcp-canada - federal, provincial, and municipal open data (StatCan, Bank of Canada, and more)
- United States:
  - bls-labor-mcp-server - Bureau of Labor Statistics (CPI, unemployment, wages, JOLTS)
  - us-census-bureau-data-api-mcp - official Census Bureau server (demographics, geography)
  - fred-mcp-server - all 800,000+ FRED time series (rates, macro indicators)
  - mcp-bea - GDP, personal income, regional accounts (early-stage: 1 commit, use with caution)
- Europe: eurostat-mcp - Eurostat statistics via the SDMX API
- Latin America: ibge-br-mcp - Brazil's IBGE geographic, demographic, and statistical data

# Usages: what about Stata?

- Stata is closed source and licensed per seat: an agent can't install it for itself the way it can `pip install` or `install.packages()`
- Command line running: the simplest option, no MCP needed
  - Run a `.do` file in batch mode: `stata-mp -b do analysis.do`
  - The agent reads the resulting `.log` file to see output and errors, then edits the `.do` file and reruns
- `stata-mcp`: a proper MCP server for richer interaction
  - Starts a real Stata session, runs commands, inspects data in memory, and extracts estimation results and graphs, not just log text
  - [github.com/hanlulong/stata-mcp](https://github.com/hanlulong/stata-mcp)

# Usages: what about Stata? (cont)

- Caveats of closed-source software with agentic AI:
  - The agent can't read Stata's internals to debug an obscure error the way it can read an R package's source on GitHub
  - Syntax changes across Stata versions (14 through 19); the agent may not know which one is actually installed on your machine
  - `.log/.smcl` output wasn't designed for machine parsing, unlike R/Python's structured objects
  - Less, and staler, Stata documentation for models to have trained on, compared to open-source tools

# Cybersecurity: local models

- Increasingly, researchers are thinking of ways on how to leverage the power of agents with restricted data
- Private datasets (e.g. tax records) contain sensitive information, and cannot be shared with external agents
- For this reason, open source models (Kimi, Llama) can potentially be installed and deployed in a local (powerful) computer
  - Most models aren't open source (e.g. Claude, ChatGPT, Gemini, Copilot), and require sending your data to the cloud
  - Open source vs. open weights: true "open source" require the weights to be available, not just the code
  - Open source models are not as powerful, and depend highly on your hardware capabilities
  - Kimi is the strongest these days, but can change quickly

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- Image: D J Shin, tin wind-up “Robby the Robot” toy, Osaka Tin Toy Institute, Wikimedia Commons (CC BY-SA 3.0)
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